

Ordinal Methods for Monocular Depth Estimation

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Abstract—Monocular depth estimation (MDE) is an important task for understanding 3D scenes. A good alternative to classical regression is using ordinal regression, which divides depth into ordered classes. However, the depth maps predicted by these models are sometimes neither smooth nor consistent.

This work studies how to add a spatial regularization term into the loss function to improve the ordinal regression models. The goal is to make the predictions smoother between neighboring pixels. We explore different strategies for this, including 1-side, 2-sides, and multi-scale regularization. We also add a small neural network to map the predicted class probabilities back to a continuous depth value, which can improve the final result.

We tested our methods on the KITTI, DIODE, and Make3D datasets using a DeepLabv3 architecture. Our results show that adding spatial regularization improves performance, especially when combined with strong ordinal loss functions like Unimodal Net and Binomial Unimodal CE. These findings show that our approach is effective for producing more accurate and visually consistent depth maps.

Index Terms—ordinal regression, monocular depth estimation, spatial regularization, deep learning, computer vision

I. INTRODUCTION

Depth estimation is crucial in many computer vision applications, providing the geometric understanding required for advanced functionality. In autonomous driving, depth estimation enables vehicles to detect and avoid obstacles, recognize traffic signs, and make safe navigation decisions by precisely measuring the distance to nearby objects. For augmented reality, it helps place virtual objects realistically, handle occlusions, and support immersive user interactions. Similarly, robotics applications depend on depth estimation for tasks such as object manipulation or environment mapping. In 3D scene reconstruction, depth estimation is essential for generating detailed models of real-world environments used in virtual reality and visualization, and special effects in film and photography. Together, these capabilities demonstrate the importance of accurate depth estimation in modern technology.

Deep learning tasks are typically categorized depending on the target: categorical or nominal tasks involve assigning data points to classes, numerical (regression) tasks involve predicting continuous values, and ordinal tasks involve predicting classes with an inherent order. While depth estimation is naturally a (truncated) continuous problem, some authors have shown it can benefit from discretization into ordered bins and taken as an ordinal regression task, e.g. Fu et al. [1]. This framing allows models to focus on relative depth ordering rather than precise value estimation, which is particularly

beneficial when geometric cues are limited, as in monocular settings.

This work focuses on monocular depth estimation (MDE), where the aim is to predict a dense depth map using only a single RGB image. Compared to stereo vision or RGB-D methods, monocular approaches are more cost-effective and hardware-independent, but also more challenging due to the absence of multiple viewpoints or direct depth sensing. To address these challenges, we explore an ordinal regression framework enhanced with spatial regularization. The proposed method encourages local consistency in the predicted depth maps while preserving structural discontinuities, resulting in more accurate and visually coherent outputs. We validate our approach on challenging outdoor datasets and analyze the impact of our contributions on depth estimation quality.



Fig. 1. RGB or monocular image and its corresponding depth map

II. RELATED WORK

Depth estimation from visual data has been extensively explored, particularly due to its significance in 3D scene understanding. This section provides an overview of the key approaches, with a focus on monocular methods and ordinal regression.

A. Stereo and RGB-D Depth Estimation

Binocular or stereo depth estimation requires two or more images taken from slightly different perspectives. The disparity (or difference in position) between corresponding points in the two images is used to triangulate the depth. While accurate

in controlled conditions, stereo methods require calibrated camera setups and struggle in textureless or occluded regions.

RGB-D depth use active sensors (e.g., Kinect, LiDAR) that capture both color and depth data. These methods are easy to use in controlled environments since it provides direct depth information, but its sensors have limitations in outdoor or large-scale environments because it must handle sensor noise, limited range, and occlusions.

B. Monocular Depth Estimation

Monocular depth estimation (MDE) only uses an ordinary camera to reconstruct depth maps based on a single image. Earlier methods relied on handcrafted features and probabilistic models like Conditional Random Fields (CRFs), which required strong scene priors and often lacked generalization.

As deep learning technologies in computer vision have advanced, deep neural networks have been applied to depth estimation tasks. Eigen et al. [2] introduced the first multi-scale convolutional neural network for monocular depth estimation laying the foundation for subsequent works. Based on this, Xie et al. [3] and others enhanced spatial precision through skip connections and encoder-decoder architectures.

Supervised deep learning models achieve high accuracy but depend on dense ground-truth depth data, which is expensive and hard to obtain at scale. To address this, unsupervised and self-supervised methods have emerged. These leverage photometric reconstruction losses from stereo images or video sequences, as seen in the work of Godard et al. [4] and Kuznetsov et al. [5]. Such approaches minimize the need for ground-truth depth while preserving competitive performance.

Beyond this, recent monocular methods introduce geometric priors or multi-scale refinements to improve depth prediction. For instance, Patil et al. [6] incorporated a planarity prior, intuitively guiding predictions in structured indoor environments by leveraging the common understanding. A key insight here is that even if two pixels on the same physical planar surface in 3D have different depths, they still share the same underlying plane equation. This allows their model anticipate and predict large, flat surfaces more accurately. Similarly, Song et al. [7] used a Laplacian pyramid decoder for coarse-to-fine detail refinement. This allows their model to first capture the overall, general depth and then progressively refine it with increasingly precise details, enhancing both large-scale structure and subtle depth variations.

C. Ordinal Regression in Monocular Depth Estimation

While many MDE approaches predict depth as a continuous variable, an alternative formulation is to treat depth estimation as an ordinal regression problem. Instead of predicting exact distances, the depth range is discretized into ordered intervals (e.g., near, mid, far), and the model learns to classify each pixel into one of these depth bins.

This method is particularly effective for monocular settings, where the lack of direct geometric cues makes precise depth prediction difficult. It improves robustness by reducing sensitivity to noise and complexity.

Fu et al. [1] introduced a pivotal approach using Spacing Increasing Discretization (SID) to map ground-truth depth into logarithmic bins. Their method, DORN, used an ordinal regression loss to learn the relative depth ordering of pixels and incorporated an ASPP-based encoder to capture multiscale spatial context.

Building on this, Li et al. [8] and Cao et al. [9] employed dilated convolutions and CRFs to refine ordinal predictions. Ochs et al. [10] further improved this approach by integrating semantic features into the depth prediction pipeline, helping the network distinguish between object boundaries and surfaces with similar textures.

Recent research focuses on strengthening ordinal MDE through self-supervision, semantic context, and transformer-based architectures, offering a promising direction for robust monocular depth estimation.

III. PROPOSAL

In this work, we investigated whether incorporating a spatial regularization term into the loss function of an ordinal regression-based monocular depth estimation network could improve prediction quality. Building on the SDNet architecture [10], which extends the DORN framework, our goal was to encourage local consistency and structural smoothness in depth predictions. However, our work produces only the depth map and not any semantic map. We explored different spatial regularization strategies to enforce consistency between neighboring pixels. The proposed method consists of the following main components:

A. Ordinal Methods – Basis

The foundation of this work was the use of ordinal regression for depth estimation. Instead of directly predicting continuous depth values, the approach discretized depth into ordered intervals and trained the network to predict these ordinal labels. This formulation is advantageous as it allows the network to capture the inherent ordering in depth values while avoiding the instability often encountered in direct regression tasks. Once the ordinal predictions are made, a post-processing step was used to map these labels back into continuous depth estimates.

B. Spatial Regularization

To further enhance the spatial smoothness and consistency of the predicted depth maps, we introduced a spatial regularization term into the loss function. The initial idea was to experiment using the term CSNP (Contact Surface Loss Using Neighbor Pixels) [11], which penalized the predicted probabilities of neighboring pixels of non-adjacent ordinal classes. The authors proposed this loss term to avoid drastic changes of semantic class between neighborhood pixels. This approach, while effective, was computationally intensive for our application. To address this, we adapted the regularization term into a more efficient form suitable for our deep ordinal regression framework. We explored three main strategies for

spatial regularization:

1) *Spatial Regularization – 1-Side*: The first approach considers only one neighboring pixel per direction. Our adapted spatial regularization term operates on the expected class predictions, computed as the weighted sum of class probabilities for each pixel to get the expected classes map,

$$\hat{y}_{i,j} = \sum_{k=0}^{K-1} k \cdot P_{k,i,j} \quad (1)$$

where:

- $\mathbf{P} \in \mathbb{R}^{K \times H \times W}$: Predicted probability tensor over K classes for an image of height H and width W ;
- $\hat{y}_{i,j}$: computes the expected class at pixel (i, j) .

This equation penalizes large differences in predicted class values between neighboring pixels, encouraging smooth transitions in the depth map. Specifically, for each pixel, we compute the absolute difference between its predicted class and that of its immediate neighbor in both the horizontal and vertical directions, respectively, only penalizing differences greater than 1 (to allow natural variations) and clamp the result at zero to ignore small differences,

$$\begin{aligned} dx_{i,j} &= \max(|\hat{y}_{i+1,j} - \hat{y}_{i,j}| - 1, 0) \\ dy_{i,j} &= \max(|\hat{y}_{i,j+1} - \hat{y}_{i,j}| - 1, 0) \end{aligned} \quad (2)$$

The mean penalty across the image is then used as the regularization term for the loss.

$$\text{CSNP}_{1\text{Side}} = \frac{1}{2} (\overline{dx} + \overline{dy}) \quad (3)$$

In Fig. 2, we illustrate the flow for computing the 1-side Spatial Regularization regularization. Starting with an input image of size $H \times W$ and K class predictions per pixel, then the expected class map is calculated by taking the weighted sum of the class probabilities. Next, we calculate the smoothness penalties along each spatial dimension. Finally, the result is mean penalty across the entire image to obtain the overall regularization value.

2) *Spatial Regularization – 2-sides*: The second variant –Equation 4– extends this idea by considering all the surrounding neighbors. It computes the weighted sum of class probabilities for each pixel in the same way. However, in this case, the penalties are calculated by getting the immediate neighbors in both directions: left and right in the horizontal axis, and top and bottom in the vertical axis. Only differences greater than 1 are also penalized, and differences are clamped at zero below this threshold.

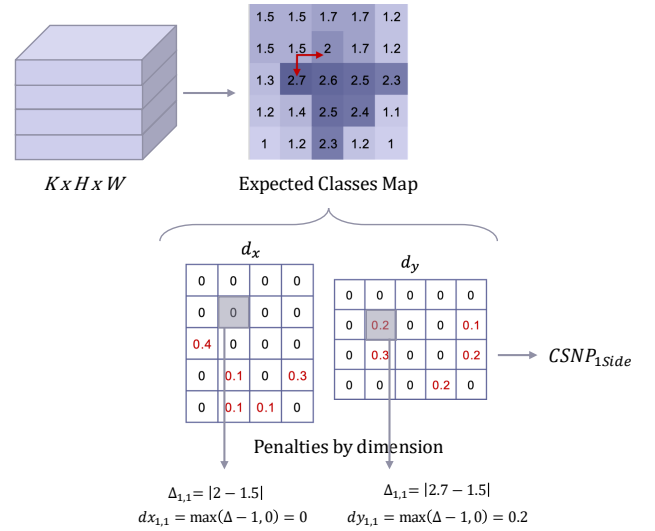


Fig. 2. Example of Spatial Regularization 1-Side calculation

$$\begin{aligned} \hat{y}_{i,j} &= \sum_{k=0}^{K-1} k \cdot P_{k,i,j} \\ dx_{i,j} &= \max(|\hat{y}_{i+1,j} - 2\hat{y}_{i,j} + \hat{y}_{i-1,j}| - 1, 0) \\ dy_{i,j} &= \max(|\hat{y}_{i,j+1} - 2\hat{y}_{i,j} + \hat{y}_{i,j-1}| - 1, 0) \\ \text{CSNP}_{2\text{Sides}} &= \frac{1}{2} (\overline{dx} + \overline{dy}) \end{aligned} \quad (4)$$

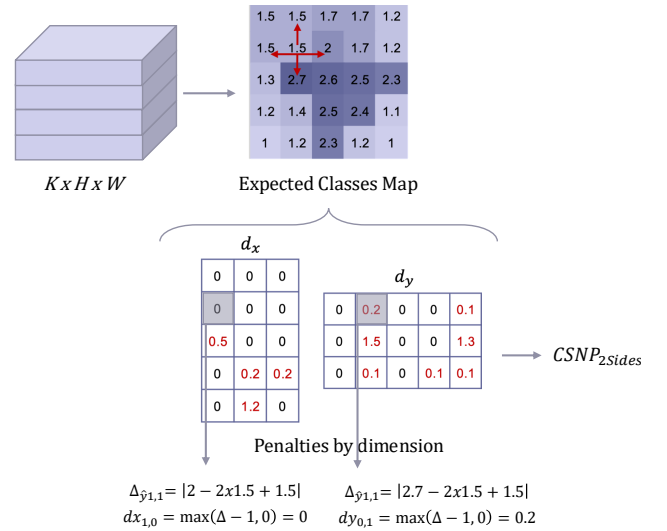


Fig. 3. Example of Spatial Regularization 2-Sides calculation

In Fig. 3, the expected class map remains the same as in the 1-Side figure. The difference lies in the calculation of the penalties, which are now computed by considering both neighboring pixels in the horizontal and vertical directions.

3) *Multi-scale Spatial Regularization*: To improve robustness, we used a multi-scale regularization strategy. In this approach, the input image and the corresponding predicted depth maps are downsampled using bilinear interpolation to $0.5\times$ and $0.25\times$ of the original resolution. Spatial regularization terms, 1-Side and 2-Sides, are then computed separately at these lower resolutions and aggregated into the final loss. This encourages the model to maintain spatial coherence not only at the original resolution but also across different scales, effectively capturing both fine and coarse structural details.

C. Continuous Depth Estimation using Probabilities

In a network that uses ordinal methods, the model produces a probability distribution across the ordinal depth bins for each pixel. To convert these bin probabilities into a continuous value, the expected value is calculated,

$$\hat{y} = \sum_{i=1}^n b_i \cdot p_i \quad (5)$$

where, n is the number of classes or bins, b is the bin's central value and p is the probability generated by the network. We also explored the Arg Max in order to compare performance. Additionally, to recover continuous depth values from the ordinal predictions, we have included a probabilistic mapping strategy in this work. After training the network using ordinal methods and applying the spatial regularization term, we implemented an additional module consisting of neural layers to learn the mapping from the probability distribution to a continuous depth value. In this approach, Equation 6, we added weights that are subsequently optimised for the gradient descent. This allows the network to leverage class confidence and interpolate between bins, resulting in smoother and more accurate depth predictions.

$$\hat{y} = \sum_{i=1}^n w_i \cdot b_i \cdot p_i \quad (6)$$

IV. EXPERIMENTAL DETAILS

Within this section, we present the details of the experimental process we followed during the development of this work.

A. Datasets

For evaluating the proposed depth estimation techniques, we conducted experiments on three widely used outdoor datasets: KITTI, Make3D, and DIODE Outdoors.

- The KITTI dataset is a benchmark suite designed for autonomous driving research. It provides high-resolution RGB images with a resolution about 375×1241 , along with corresponding LiDAR-based depth maps. All the data was collected in urban environments using a vehicle equipped with stereo cameras, a 3D laser scanner, and a GPS/IMU system. The dataset supports a range of vision tasks such as stereo matching, optical flow, visual

odometry, and object detection, making it highly versatile for evaluating computer vision algorithms under realistic driving conditions. We followed the split used by Eigen et al. [2] with a total of 44,506 images for training, 1,744 for validation, and 649 as testing dataset.

- The Make3D dataset contains 534 pairs of outdoor RGB and depth images, with 400 images for training (of which 40 were held out for validation) and 134 for testing. The RGB images are high-resolution with 2272×1704 , while the depth maps, obtained via a laser scanner, are lower in resolution with 55×305 . Make3D covers a variety of outdoor scenes, including cityscapes and natural landscapes, and was one of the earliest datasets to spur development in monocular depth estimation.
- The DIODE dataset (Dense Indoor and Outdoor Depth Estimation) provides high-quality with a resolution of 1024×768 , dense depth maps for both indoor and outdoor scenes. For our experiments, we used the outdoor subset, which offers challenging and diverse real-world scenes with accurate ground-truth depth, making it valuable for benchmarking depth estimation models in complex outdoor environments. The split for the image datasets is: 15,196 images for training, 1,688 for the validation and 446 images for testing.

B. Network Architecture

In our experiments, we used the DeepLab network, motivated by its successful application in several monocular depth estimation works. Specifically, we implemented DeepLabv3 with a ResNet-101 backbone pre-trained on ImageNet to extract rich, hierarchical features from input images. For our task, we adapted the final classifier layer to output depth discretization classes. To stabilize training, all batch normalization layers were fixed in evaluation mode.

In one of our experiments (Section III-C), we aimed to predict continuous depth values directly from the ordinal probability outputs of the DeepLab network. To achieve this, we introduced an OrdinalMapping module that takes the vector of per-class probabilities, produced by the ordinal classification head, and maps it to a single depth estimate per pixel. This module consists of three fully connected layers with dimensions 128, 64, and 128, respectively. The final layer is followed by a softplus activation to ensure that the predicted depth values are positive and smoothly distributed.

C. Losses

In our experiments, we evaluated several loss functions tailored for monocular depth estimation with discretized depth intervals. The standard Cross-Entropy (CE) loss serves as a baseline, considering depth intervals as independent classes without considering their internal order. To address this, Ordinal Coding and Class Distance Weighted Cross-Entropy (CDW CE) introduce ordinal awareness: Ordinal Coding reformulates the problem as a series of ordered binary decisions, while CDW CE penalizes misclassifications proportionally to the

distance between the predicted and true intervals. The Unimodal Network and Unimodal Binomial CE further improve prediction quality by enforcing the network to produce unimodal (single-peaked) probability distributions, reflecting the natural structure of depth. Finally, the CO2 loss encourages ordinal consistency in the output by penalizing violations of the expected ordering among depth intervals. Together, these losses leverage the ordinal and unimodal properties of depth, resulting in more accurate and consistent depth predictions. In experiments with Spatial Regularization, the computed term is simply added to the loss before backpropagation.

D. Spatial Regularization

As we mentioned in the Proposal section, in our experiments, we incorporated spatial regularization into our model's loss function. This technique helps improve the quality of predicted depth maps by encouraging spatial smoothness.

We explored two methods for applying spatial regularization:

- Spatial Regularization (1-Side): This approach focuses on the immediate difference in predicted depth between a pixel and its direct horizontal and vertical neighbors.
- Spatial Regularization (2-Sides): This extends the regularization by considering a broader neighborhood, looking at both horizontal and vertical neighbors on both sides.

Both of these regularization terms are multiplied by a λ value before being added to the overall loss function. This λ value acts as a weight, allowing us to control how strongly the spatial smoothness is enforced.

Our goal with these regularization terms was to encourage the network to generate depth maps that are not only accurate but also visually consistent. While effective in achieving smoother results, it's worth noting that applying these regularization terms did increase the computational resources needed during training.

Finally, we adopted a multi-scale approach. This means we applied both the 1-Side and 2-Sides spatial regularization independently at different scales of the predicted depth maps. This allowed for a more nuanced and robust application of the smoothing effect across various levels of detail in our predictions.

E. Discretizer

We used the Spacing-Increasing Discretization (SID) strategy, introduced in DORN by Fu et al. [1]. SID divides the continuous depth range into intervals that grow exponentially with distance, Fig. 4, allowing for finer resolution for nearby depths and coarser resolution for deeper depths. This is consistent with the observation that depth estimation errors are most critical at close range and less noticeable at greater distances. By applying SID, the network can better focus on accurately predicting small and medium depths while still providing reasonable estimates for greater depths, ultimately improving depth estimation performance compared to linear discretization.



Fig. 4. SID to discretize depth interval into five subintervals

F. Data Augmentation

To improve the robustness and generalization of our depth estimation models, we applied a set of data augmentation techniques tailored to each dataset. For the KITTI dataset, we performed a slight upscaling followed by random cropping to 640 x 192, horizontal flipping with 50% probability, and adaptive histogram equalization (CLAHE) to enhance local contrast. For DIODE and Make3D, all images were resized to a resolution of 512 x 384, and random horizontal flipping was applied; additionally, DIODE included a channel shuffle augmentation with a 10% probability to increase color diversity. For all datasets, images were normalized using the standard mean and standard deviation values from ImageNet, and finally converted to tensors for model input.

G. Training Methodology and Setup

The training of the proposed depth estimation model was carried out in the following experimental setup. The network was trained for 10 epochs with a batch size of 8. The number of ordinal classes was set to 128 to discretize the depth range, this number was adopted from the SDNet [10] where the authors explain that using that number of bins got the best results. For optimization, the AdamW optimizer was used with a learning rate of 1×10^{-4} and a weight decay parameter of 1×10^{-4} to promote regularization and prevent overfitting.

Regarding the incorporation of the spatial regularization term into the loss function, multiple weighting coefficients (λ) were experimented with, specifically: 0.001, 0.01, 100, and 10,000. However, the best results were obtained for $\lambda = 0.001$, so only these results are present in the tables, with the rest omitted from this report.

Furthermore, all experiments were performed on a virtual machine equipped with an NVIDIA A100-SXM4-40GB GPU, provided by the ATLAS research group at the Faculty of Engineering, University of Porto.

H. Evaluation Metrics

To evaluate the performance of the proposed depth estimation methods, we employed error-based metrics commonly used in monocular depth estimation. Each metric provided complementary insights into error magnitude and robustness across depth ranges.

Let N be the total number of pixels in the evaluation set, d_i the ground truth depth at pixel i , and $\hat{d}_i$ the predicted depth at the same pixel. The metrics used were the following:

1) Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |d_i - \hat{d}_i| \quad (7)$$

Measures the average absolute difference between predicted and true depth values.

2) *Mean Squared Error (MSE)*:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (d_i - \hat{d}_i)^2 \quad (8)$$

Penalizes larger errors more than MAE by squaring the difference.

3) *Absolute Relative Error (AbsRel)*:

$$\text{AbsRel} = \frac{1}{N} \sum_{i=1}^N \frac{|d_i - \hat{d}_i|}{d_i} \quad (9)$$

Normalizes the absolute error by the ground truth value, offering a scale-invariant measure.

4) *Squared Relative Error (SqRel)*:

$$\text{SqRel} = \frac{1}{N} \sum_{i=1}^N \frac{(d_i - \hat{d}_i)^2}{d_i} \quad (10)$$

Similar to AbsRel but gives higher weight to larger errors through squaring.

5) *Root Mean Squared Log Error (RMSELog)*:

$$\text{RMSELog} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\log d_i - \log \hat{d}_i)^2} \quad (11)$$

Computes the RMSE in log space to treat under- and over-estimations symmetrically across depth ranges.

V. RESULTS AND DISCUSSION

For each dataset evaluated –KITTI, DIODE, and Make3D– the experiments are presented in two different tables. The first table in each case provides a comparison of various ordinal methods with spatial regularization, differentiating between two predicting strategies: Arg Max and Expected Value, summarizing the performance using the metrics explained in the section above. The second table extends the analysis by evaluating the impact of applying mapping for continuous depth prediction.

In Table I, we observe that KITTI results generally improved performance over the base settings when spatial regularization was used. Among the methods, Unimodal Net and Binomial Unimodal CE shown the best results, with Unimodal Net achieving the lowest MAE=1.95 (Arg Max and Expected Value) and absRel=11.87 (Arg Max) when combined with spatial regularization multi-scale. However, the most significant improvements are observed in Table II, which focuses on the impact of mapping and further use of spatial regularization. Here, all the loss functions demonstrate enhanced performance when mapping is incorporated, with Unimodal Net achieving the best metrics, specifically, MAE=1.91, an AbsRel=11.54 and a SqRel=2.95. When combined with spatial regularisation (1-Side), the MSE was further reduced to 13.63. These results highlight that mapping, especially when combined with spatial

regularization, is effective for this dataset, and Unimodal Net stands out as the most robust approach.

On the DIODE dataset –results in Table III– spatial regularization consistently improves depth estimation performance for most loss functions, with Binomial Unimodal CE, Ordinal Encoding, and Unimodal Net showing strong results. For example, Binomial Unimodal CE with multi-scale spatial regularization achieved of MAE=5.19 (Arg Max) and MAE=5.38 (Expected Value) and AbsRel=46.42 (Arg Max), while Ordinal Encoding with 1-Side regularization reached MAE=4.87 and AbsRel=45.06 in Arg Max, outperforming other base configurations. The introduction of ordinal mapping, results in Table IV, as in the Kitti dataset further enhanced the ordinal method results, with Binomial Unimodal CE, Ordinal Encoding and Unimodal Net achieving their best metrics in combination with mapping and multi-scale spatial regularization (2-Sides), such as MAE=4.91 and AbsRel=43.20 for Binomial Unimodal CE, MAE=4.63 and AbsRel=43.78 for Ordinal Encoding and, finally, Unimodal Net with MAE=4.84 and AbsRel=46.11.

On the Make3D dataset, Table V, the trends observed in KITTI and DIODE persisted, where spatial regularization generally led to modest improvements across most loss functions, with the best results typically observed when using 2-Sides or multi-scale regularization. For example, the Binomial Unimodal CE with 2-Sides regularization achieved a MAE=6.31 (Arg Max) and 6.35 (Expected Value), with the lowest AbsRel (28.68 and 28.61, respectively), while UnimodalNet and OrdinalEncoding also showed improved accuracy when spatial regularization was applied. However, it is noticeable and important to mention that, unlike the Kitti and DIODE datasets, applying mapping did not improve the results for this dataset, probably because this one is too small for the network to learn properly.

In general, across all datasets, spatial regularization improves depth estimation performance, with the best results observed when it is applied for 2-Sides or multi-scale configurations. The use of the mapping usually provides further enhancements, particularly when combined with spatial regularization, leading to more accurate results. Among the various loss functions evaluated, Binomial Unimodal CE and Unimodal Net consistently demonstrate the most robust and effective performance across different datasets and experimental settings. Although the metrics vary depending on the characteristics of each dataset, the relative improvements achieved through the application of regularization and mapping remain consistent, highlighting the generalizability of these strategies.

VI. CONCLUSION

In this work, we explored the effectiveness of various ordinal methods and spatial regularization techniques for depth estimation in outdoor datasets. Our results showcased the improvements achieved with the combination of spatial regularization with robust ordinal loss functions, such as Binomial

Unimodal CE and Unimodal Net, particularly in the Kitti dataset. By leveraging multi-scale spatial regularization, our models were able to maintain spatial coherence at multiple resolutions, optimizing both fine and coarse details. Additionally, incorporating a mapping strategy for continuous depth estimation, further improves prediction smoothness and accuracy. These findings highlight the potential of our approach to enhance depth mapping capabilities. Future work may involve exploring these techniques in more complex and diverse environments to further validate and expand their applicability.

Additionally, monocular depth estimation using neural networks contributes to several United Nations Sustainable Development Goals [12], enabling low-cost, efficient perception systems that support a wide range of applications. In industry and infrastructure (Goal 9), it drives innovation in robotics, AR/VR, and intelligent monitoring systems. For sustainable cities (Goal 11), it aids autonomous mobility, urban planning, and public safety. In climate action (Goal 13), it enables lightweight, camera-based environmental monitoring using drones.

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TABLE I
KITTI DATASET – COMPARISON OF LOSS FUNCTIONS AND SPATIAL REGULARIZATION METHODS

Loss name	Setting	Arg Max					Expected Value				
		MAE	MSE	absRel	sqRel	RMSELog	MAE	MSE	absRel	sqRel	RMSELog
BinomialUnimodal CE	Base	2.06	14.34	12.94	3.26	0.16	2.16	14.55	14.14	3.61	0.17
+ SpatialReg ($\lambda = 0.001$)	1Side	2.01	14.35	12.61	3.18	0.16	2.12	14.54	13.79	3.52	0.16
	2Sides	2.06	14.24	13.14	3.37	0.16	2.16	14.48	14.37	3.74	0.17
+ SpatialRegMulti	1Side	2.00	14.36	12.36	3.15	0.16	2.10	14.56	13.50	3.48	0.16
	2Sides	2.02	14.24	12.56	3.20	0.16	2.12	14.65	13.73	3.53	0.16
CDW CE	Base	2.39	22.57	14.01	3.96	0.18	2.35	21.42	13.87	3.83	0.17
+ SpatialReg ($\lambda = 0.001$)	1Side	2.27	17.39	13.96	3.73	0.17	2.19	16.25	13.63	3.55	0.17
	2Sides	2.25	17.31	13.84	3.70	0.17	2.20	16.53	13.60	3.56	0.17
+ SpatialRegMulti	1Side	2.29	17.97	13.92	3.75	0.17	2.19	16.60	13.43	3.52	0.17
	2Sides	2.32	18.26	14.16	3.90	0.17	2.20	16.53	13.69	3.65	0.17
CE	Base	2.06	18.02	12.36	3.88	0.17	2.04	15.71	12.68	4.30	0.16
+ SpatialReg ($\lambda = 0.001$)	1Side	2.10	18.25	12.76	3.93	0.17	2.11	15.99	13.39	4.49	0.17
	2Sides	2.07	18.11	12.39	3.65	0.17	2.06	15.72	12.89	3.97	0.16
+ SpatialRegMulti	1Side	2.05	17.23	12.47	3.49	0.17	2.05	15.01	13.15	4.05	0.17
	2Sides	2.08	18.25	12.50	3.79	0.17	2.09	15.77	13.34	4.48	0.17
CO2	Base	2.04	17.56	12.22	3.49	0.17	2.01	15.20	12.45	3.69	0.16
+ SpatialReg ($\lambda = 0.001$)	1Side	2.08	17.55	12.53	3.58	0.17	2.08	15.49	13.18	4.32	0.17
	2Sides	2.07	17.55	12.43	3.55	0.17	2.07	15.52	13.02	4.04	0.17
+ SpatialRegMulti	1Side	2.07	17.56	12.52	3.62	0.17	2.07	15.24	13.19	4.08	0.17
	2Sides	2.08	17.71	12.63	3.65	0.17	2.09	15.41	13.50	4.49	0.17
OrdinalEncoding	Base	2.12	17.21	13.08	3.95	0.17	2.10	15.51	13.22	3.75	0.16
+ SpatialReg ($\lambda = 0.001$)	1Side	2.09	17.69	12.68	4.12	0.17	2.07	15.35	13.11	3.95	0.17
	2Sides	2.04	16.53	12.57	3.82	0.16	2.03	14.93	12.72	3.54	0.16
+ SpatialRegMulti	1Side	2.13	17.86	13.32	5.83	0.17	2.24	17.18	15.58	8.49	0.20
	2Sides	2.14	17.74	13.25	4.73	0.17	2.13	15.17	13.80	4.17	0.17
UnimodalNet	Base	2.04	15.70	12.62	3.49	0.16	2.03	14.54	12.71	3.41	0.16
+ SpatialReg ($\lambda = 0.001$)	1Side	1.96	14.44	12.05	3.00	0.16	1.96	13.92	12.11	2.95	0.16
	2Sides	1.96	14.68	12.07	3.26	0.16	1.96	13.91	12.16	3.20	0.16
+ SpatialRegMulti	1Side	1.95	14.67	11.87	3.00	0.16	1.95	14.07	11.91	2.94	0.15
	2Sides	2.00	14.95	12.33	3.21	0.16	2.00	14.53	12.37	3.14	0.16

TABLE II
KITTI DATASET – COMPARISON OF ORDINAL METHODS + MAPPING AND SPATIAL REGULARIZATION METHODS

Loss name	Setting	MAE	MSE	absRel	sqRel	RMSELog
BinomialUnimodal CE	Base	2.06	14.34	12.94	3.26	0.16
SpatialReg ($\lambda = 0.001$)	Mapping	2.01	14.11	12.54	3.15	0.16
	1Side	2.01	14.39	12.49	3.20	0.16
+ SpatialRegMulti	2Sides	1.99	14.12	12.36	3.20	0.16
	1Side	1.99	14.42	12.23	3.19	0.16
	2Sides	1.97	14.34	12.12	3.11	0.15
CDW CE	Base	2.39	22.57	14.01	3.96	0.18
SpatialReg ($\lambda = 0.001$)	Mapping	2.10	16.14	12.73	3.38	0.16
	1Side	2.10	15.38	13.04	3.38	0.16
+ SpatialRegMulti	2Sides	2.11	15.76	12.82	3.38	0.16
	1Side	2.12	15.98	12.98	3.45	0.16
	2Sides	2.10	16.04	12.65	3.38	0.16
CE	Base	2.06	18.02	12.36	3.88	0.17
SpatialReg ($\lambda = 0.001$)	Mapping	2.01	15.34	12.36	3.51	0.16
	1Side	2.02	15.44	12.47	3.66	0.16
+ SpatialRegMulti	2Sides	2.01	15.34	12.36	3.51	0.16
	1Side	1.98	14.77	12.20	3.48	0.16
	2Sides	2.01	15.02	12.55	3.68	0.16
CO2	Base	2.04	17.56	12.22	3.49	0.17
SpatialReg ($\lambda = 0.001$)	Mapping	1.99	14.99	12.36	3.62	0.16
	1Side	2.04	15.46	12.57	3.63	0.16
+ SpatialRegMulti	2Sides	1.98	15.22	12.04	3.43	0.16
	1Side	2.00	14.89	12.34	3.44	0.16
	2Sides	2.00	14.99	12.36	3.45	0.16
OrdinalEncoding	Base	2.12	17.21	13.08	3.95	0.17
SpatialReg ($\lambda = 0.001$)	Mapping	1.99	14.89	12.14	3.27	0.16
	1Side	1.98	15.20	11.91	3.20	0.16
+ SpatialRegMulti	2Sides	1.96	14.85	12.04	3.27	0.16
	1Side	2.01	15.27	12.40	3.44	0.16
	2Sides	1.98	14.93	12.33	3.38	0.16
UnimodalNet	Base	2.04	15.70	12.62	3.49	0.16
SpatialReg ($\lambda = 0.001$)	Mapping	1.91	14.12	11.54	2.95	0.15
	1Side	1.92	13.63	11.78	2.97	0.15
+ SpatialRegMulti	2Sides	1.96	13.95	12.20	3.22	0.16
	1Side	1.95	13.97	11.99	3.03	0.15
	2Sides	1.94	13.96	11.62	3.00	0.15

TABLE III
DIODE DATASET – COMPARISON OF ORDINAL METHODS AND SPATIAL REGULARIZATION METHODS

Loss name	Setting	Arg Max					Expected Value				
		MAE	MSE	absRel	sqRel	RMSELog	MAE	MSE	absRel	sqRel	RMSELog
BinomialUnimodal CE	Base	5.76	100.27	51.89	131.69	0.52	6.01	104.55	54.92	142.56	0.53
+ SpatialReg ($\lambda = 0.001$)	1Side	5.45	92.30	50.08	123.95	0.51	5.67	95.60	53.01	134.32	0.52
	2Sides	5.94	100.11	54.95	142.73	0.53	6.22	104.87	58.31	154.65	0.54
+ SpatialRegMulti	1Side	5.19	89.90	46.42	110.63	0.49	5.38	92.49	49.16	119.84	0.50
	2Sides	5.25	90.02	46.43	113.84	0.49	5.45	92.91	49.17	123.12	0.50
CDW CE	Base	5.65	105.51	46.72	76.36	0.52	5.65	105.85	47.12	76.94	0.52
+ SpatialReg ($\lambda = 0.001$)	1Side	5.67	106.06	46.49	81.02	0.53	5.66	106.41	46.87	81.63	0.53
	2Sides	5.71	107.10	46.71	80.23	0.53	5.68	106.64	47.11	80.73	0.52
+ SpatialRegMulti	1Side	5.71	106.78	45.37	70.40	0.52	5.70	107.07	45.50	70.33	0.52
	2Sides	5.63	107.61	43.38	66.51	0.52	5.60	107.63	43.62	66.89	0.51
CE	Base	6.11	116.46	56.61	200.97	0.59	5.28	87.26	54.57	184.72	0.53
+ SpatialReg ($\lambda = 0.001$)	1Side	5.77	109.81	53.11	184.21	0.57	7.75	128.91	95.45	349.50	0.71
	2Sides	6.16	116.73	56.91	192.86	0.59	6.07	98.40	68.31	228.53	0.59
+ SpatialRegMulti	1Side	5.98	113.05	56.39	199.04	0.58	8.61	148.26	113.90	537.01	0.78
	2Sides	6.13	117.10	57.51	203.07	0.59	9.26	161.65	120.96	548.93	0.81
CO2	Base	5.84	114.31	53.34	194.56	0.57	5.13	87.26	52.67	184.14	0.52
+ SpatialReg ($\lambda = 0.001$)	1Side	5.93	113.24	55.06	191.76	0.57	7.79	129.36	95.56	343.12	0.71
	2Sides	6.02	115.81	54.54	189.29	0.58	5.92	97.13	66.28	230.59	0.58
+ SpatialRegMulti	1Side	5.90	114.78	55.04	199.76	0.57	8.24	142.05	106.36	494.07	0.75
	2Sides	5.91	113.30	53.45	176.06	0.58	8.87	153.47	114.54	511.55	0.78
OrdinalEncoding	Base	5.07	93.77	47.44	175.96	0.49	4.93	79.19	49.77	152.08	0.49
+ SpatialReg ($\lambda = 0.001$)	1Side	4.87	88.41	45.06	144.82	0.48	4.90	77.37	50.94	144.55	0.50
	2Sides	5.03	90.47	46.35	155.97	0.49	4.90	78.13	49.16	146.65	0.49
+ SpatialRegMulti	1Side	5.02	89.55	46.15	143.17	0.49	5.48	83.33	62.20	173.61	0.55
	2Sides	4.91	88.91	45.26	146.47	0.48	5.03	79.52	53.19	153.54	0.51
UnimodalNet	Base	5.69	103.02	54.42	192.53	0.54	5.30	90.10	52.06	158.83	0.51
+ SpatialReg ($\lambda = 0.001$)	1Side	5.63	100.82	54.78	193.02	0.53	5.32	89.99	52.91	163.90	0.52
	2Sides	5.49	97.38	52.91	182.21	0.53	5.15	85.43	50.88	150.81	0.51

TABLE IV
DIODE DATASET – COMPARISON OF ORDINAL METHODS + MAPPING AND SPATIAL REGULARIZATION

Loss name	Setting	MAE	MSE	absRel	sqRel	RMSELog
BinomialUnimodal CE	Base	5.76	100.27	51.89	131.69	0.52
SpatialReg ($\lambda = 0.001$)	Mapping	5.42	87.96	50.26	116.30	0.51
	1Side	5.08	84.96	46.83	108.24	0.50
+ SpatialRegMulti	2Sides	5.34	87.87	50.36	121.01	0.51
	1Side	4.93	84.02	43.81	99.24	0.48
CDW CE	2Sides	4.91	83.19	43.20	98.65	0.48
	Base	5.65	105.51	46.72	76.36	0.52
SpatialReg ($\lambda = 0.001$)	Mapping	6.01	98.22	56.78	125.25	0.54
	1Side	6.24	105.44	61.68	155.86	0.57
+ SpatialRegMulti	2Sides	6.14	104.14	58.70	133.39	0.56
	1Side	5.94	97.50	55.90	116.86	0.54
CE	2Sides	5.95	100.88	55.15	120.37	0.54
	Base	6.11	116.46	56.61	200.97	0.59
SpatialReg ($\lambda = 0.001$)	Mapping	5.25	87.96	52.89	155.76	0.52
	1Side	5.19	86.56	53.18	156.59	0.52
+ SpatialRegMulti	2Sides	5.26	88.01	52.09	152.79	0.52
	1Side	5.09	82.76	51.10	144.12	0.51
CO2	2Sides	5.27	86.02	54.00	151.01	0.52
	Base	5.84	114.31	53.34	194.56	0.57
SpatialReg ($\lambda = 0.001$)	Mapping	5.22	86.83	54.18	153.32	0.52
	1Side	5.49	91.57	57.10	172.09	0.53
+ SpatialRegMulti	2Sides	5.08	85.52	50.12	150.31	0.50
	1Side	5.10	85.34	51.39	154.32	0.51
OrdinalEncoding	2Sides	5.12	85.07	51.39	143.76	0.51
	Base	5.07	93.77	47.44	175.96	0.49
SpatialReg ($\lambda = 0.001$)	Mapping	4.72	77.31	44.58	119.01	0.48
	1Side	4.63	74.55	44.33	117.01	0.47
+ SpatialRegMulti	2Sides	4.92	81.45	46.40	128.14	0.48
	1Side	4.71	77.26	44.54	118.65	0.47
UnimodalNet	2Sides	4.63	76.20	43.78	118.84	0.46
	Base	5.69	103.02	54.42	192.53	0.54
SpatialReg ($\lambda = 0.001$)	Mapping	4.96	85.08	47.82	141.77	0.50
	1Side	5.00	85.04	48.42	143.59	0.50
+ SpatialRegMulti	2Sides	4.84	81.12	46.11	131.61	0.49
	1Side	4.93	84.69	45.73	137.71	0.49
	2Sides	4.94	84.89	46.80	139.23	0.49

TABLE V
MAKE3D DATASET – COMPARISON OF ORDINAL METHODS AND SPATIAL REGULARIZATION

Loss name	Setting	Arg Max					Expected Value				
		MAE	MSE	absRel	sqRel	RMSELog	MAE	MSE	absRel	sqRel	RMSELog
BinomialUnimodal CE	Base	6.35	188.36	28.68	32.19	0.43	6.39	188.60	28.61	32.47	0.43
+ SpatialReg ($\lambda = 0.001$)	1Side	6.61	196.28	30.70	31.86	0.45	6.64	196.45	30.38	31.98	0.45
	2Sides	6.31	186.64	28.76	33.71	0.43	6.35	186.87	28.75	34.01	0.43
+ SpatialRegMulti	1Side	6.53	194.71	30.09	31.20	0.44	6.56	194.94	29.79	31.36	0.44
	2Sides	6.48	191.39	30.44	34.35	0.44	6.50	191.60	30.15	34.50	0.44
CDW CE	Base	16.32	584.71	109.89	395.00	0.88	12.75	309.42	104.04	203.40	0.76
+ SpatialReg ($\lambda = 0.001$)	1Side	16.29	583.56	109.80	398.71	0.88	12.73	307.55	104.04	204.53	0.76
	2Sides	16.21	577.45	109.60	397.66	0.88	12.61	304.28	103.42	205.21	0.76
+ SpatialRegMulti	1Side	16.59	595.84	113.65	419.58	0.89	12.71	307.44	104.14	209.48	0.76
	2Sides	16.18	577.35	109.13	396.57	0.87	12.61	303.34	103.25	202.35	0.76
CE	Base	10.71	529.49	77.13	577.88	0.67	10.54	257.01	100.60	224.68	0.78
+ SpatialReg ($\lambda = 0.001$)	1Side	10.75	531.69	77.28	587.04	0.67	10.60	259.20	101.04	226.11	0.78
	2Sides	11.35	579.55	82.79	640.94	0.69	10.43	248.70	98.10	220.02	0.76
+ SpatialRegMulti	1Side	12.01	620.04	87.05	649.52	0.71	10.86	255.28	105.34	245.31	0.79
	2Sides	12.03	622.35	87.38	658.97	0.71	10.57	248.66	101.27	230.44	0.77
CO2	Base	11.57	594.20	84.99	661.29	0.69	10.56	251.64	99.21	224.99	0.76
+ SpatialReg ($\lambda = 0.001$)	1Side	10.87	540.49	78.65	598.74	0.67	10.51	253.74	100.88	225.97	0.78
	2Sides	11.33	577.89	82.42	625.63	0.69	10.49	252.25	98.10	221.12	0.76
+ SpatialRegMulti	1Side	12.05	623.81	88.19	680.67	0.71	10.67	245.94	105.45	249.07	0.79
	2Sides	11.98	620.24	87.05	658.27	0.70	10.58	247.60	101.47	231.87	0.77
OrdinalEncoding	Base	14.98	597.05	109.50	219.53	0.94	17.17	611.93	158.54	470.67	1.08
+ SpatialReg ($\lambda = 0.001$)	1Side	14.17	586.08	108.67	204.10	0.93	17.16	611.65	158.41	469.59	1.08
	2Sides	14.18	586.06	108.69	224.10	0.93	17.17	611.52	158.37	468.54	1.08
+ SpatialRegMulti	1Side	14.98	596.23	109.07	219.53	0.94	17.16	610.93	158.65	471.68	1.08
	2Sides	14.17	584.29	102.15	205.15	0.92	17.15	611.11	158.03	466.89	1.08
UnimodalNet	Base	11.80	656.14	30.69	23.15	0.78	11.80	657.02	30.72	23.23	0.78
+ SpatialReg ($\lambda = 0.001$)	1Side	11.84	657.75	31.03	24.24	0.78	11.84	658.71	31.05	24.33	0.78
	2Sides	11.79	655.57	31.06	24.70	0.78	11.79	656.36	31.09	24.79	0.78
+ SpatialRegMulti	1Side	11.57	635.14	29.91	22.79	0.74	11.56	635.94	29.89	22.87	0.75
	2Sides	11.60	637.49	30.20	22.94	0.75	11.60	638.51	30.17	23.01	0.75

TABLE VI
MAKE3D DATASET – COMPARISON OF ORDINAL METHODS + MAPPING AND SPATIAL REGULARIZATION

Loss name	Setting	MAE	MSE	absRel	sqRel	RMSELog
BinomialUnimodal	Base	6.35	188.36	28.68	32.19	0.43
SpatialReg ($\lambda = 0.001$)	Mapping	12.48	489.97	73.91	95.70	0.73
	1Side	12.47	491.68	73.60	95.77	0.73
+ SpatialRegMulti	2Sides	12.48	489.98	73.93	95.74	0.73
	1Side	12.46	491.38	73.40	95.02	0.73
	2Sides	12.47	491.12	73.78	97.49	0.73
CDW	Base	16.32	584.71	109.89	395.00	0.88
SpatialReg ($\lambda = 0.001$)	Mapping	13.89	580.81	91.61	154.75	0.88
	1Side	13.88	579.47	91.94	156.02	0.88
+ SpatialRegMulti	2Sides	13.89	580.79	91.61	154.75	0.88
	1Side	13.81	578.00	90.55	151.20	0.87
	2Sides	13.97	582.28	93.00	159.47	0.88
CE	Base	10.71	529.49	77.13	577.88	0.67
SpatialReg ($\lambda = 0.001$)	Mapping	14.95	605.61	109.14	224.51	0.95
	1Side	15.05	608.51	110.64	230.64	0.95
+ SpatialRegMulti	2Sides	14.95	605.60	109.14	224.51	0.95
	1Side	14.98	606.11	109.58	226.45	0.95
	2Sides	14.93	605.27	108.76	222.90	0.95
CO2	Base	11.57	594.20	84.99	661.29	0.69
SpatialReg ($\lambda = 0.001$)	Mapping	14.97	605.58	109.49	226.05	0.95
	1Side	15.01	608.09	109.97	227.80	0.95
+ SpatialRegMulti	2Sides	14.97	605.58	109.49	226.04	0.95
	1Side	14.93	604.61	108.88	223.58	0.95
	2Sides	14.95	604.31	109.24	224.96	0.95
OrdinalEncoding	Base	14.98	597.05	110.87	225.82	0.95
SpatialReg ($\lambda = 0.001$)	Mapping	14.89	596.88	109.50	219.52	0.94
	1Side	14.91	595.84	109.94	220.91	0.94
+ SpatialRegMulti	2Sides	14.89	596.87	109.50	219.53	0.94
	1Side	14.95	595.72	110.91	225.73	0.94
	2Sides	14.92	595.07	110.39	222.82	0.94
UnimodalNet	Base	11.80	656.14	30.69	23.15	0.78
SpatialReg ($\lambda = 0.001$)	Mapping	13.77	539.96	82.70	119.68	0.79
	1Side	13.74	540.09	82.23	117.44	0.79
+ SpatialRegMulti	2Sides	13.77	539.95	82.70	119.69	0.79
	1Side	13.23	528.01	74.86	95.69	0.75
	2Sides	13.35	533.07	76.43	100.27	0.76